Clariant Produkte (Deutschland) GmbH
Analytical Sciences, G860
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REPORT OF ANALYSIS - PHYSICS, PARTICLE & SURFACE ANALYSIS

Sample:	**TB1180A**
Order title/Product:	Graphene / Raman microscopy
Project:	Graphene
Account:	Curia Germany GmbH
Customer:	Dr. Thorsten Busch
Order no.:	25-050404-1
Remarks:	Raman measurement on request of the customer

Method:

Raman spectroscopy/microscopy using Senterra I Raman microscope (Bruker).

Procedure:

Objective:	Raman spectroscopy
Field of Application:	Powder
Principle:	The substance is applied to a quarz microscope slide and examined under the microscope. The objective is selected and focused to examine an exact location using Raman. At this position, several points are examined with the laser and a corresponding number of Raman spectra are recorded.
Reference:	N/A

Experimental Details and Result

Equipment:	Senterra I Raman microscope (Bruker)
Magnification:	20 x
Wavenumber range:	45 – 3700 cm^{-1}
Wavenumber step size	3 – 5 cm^{-1}
Co-addition:	100
Exposure time:	10000 ms
Evaluation:	Averaging over similar spectra

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Result:

Absorption band	2D band	G band	D band	D/G ratio
Integration range	2791 - 2557	1753,2 – 1474	1474 – 959	
Integral Typ A	2,581	228,0	330,8	1,451
Peak high Typ J	0,024	1,972	1,784	0,905

Spectra see attachment.

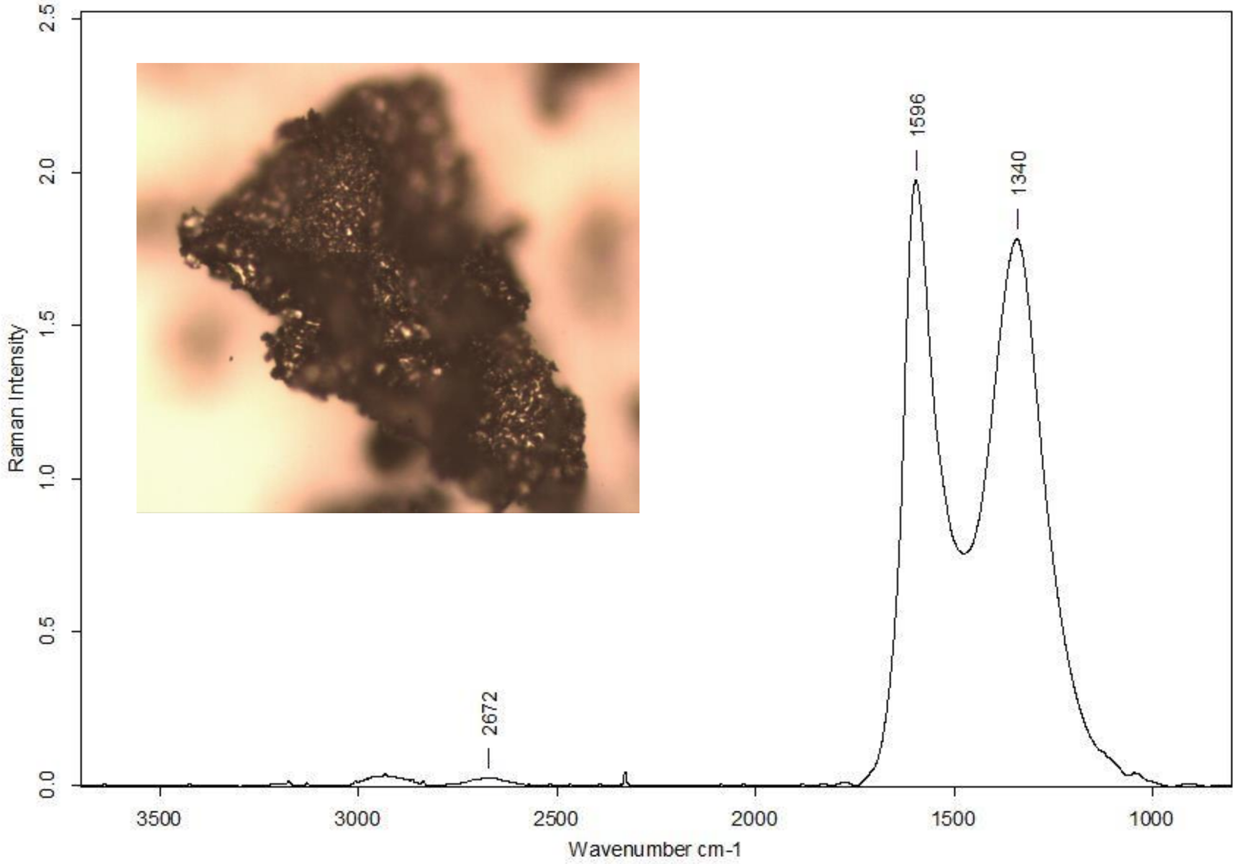

Figure 1: TB1180A, average spectrum over 8 recordings

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